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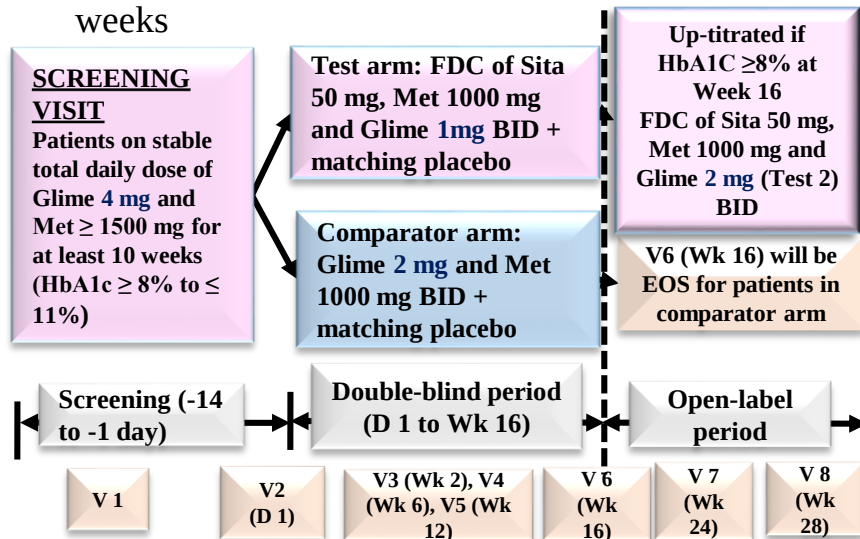
BACKGROUND

Sitagliptin addition to metformin and a lower dose of glimepiride:

- The prolonged activity of intact GLP-1 by DPP-4 inhibition is likely to both improve glucose-induced insulin secretion and to decrease glucagon levels, resulting in the maintenance of β -cell function¹
- Reduced hypoglycemia events and is weight neutral
- Provides a complementary mechanism for glucose induced insulin secretion leading to improved diabetes control
- Sitagliptin exhibits known synergism with Metformin through several mechanisms²

METHODS AND STUDY DESIGN

- This is a subgroup analysis of a phase 3, randomized, double-blind, double-dummy, active controlled study (16 weeks) followed by an open-label, single-arm study (12 weeks) [CTRI/2021/11/038169]
- The study included adult Indian patients who were randomized to receive Test OR Comparator for 16 weeks



BID: Twice daily; D: Day; EOS: End of study; FDC: Fixed dose combination; Glime; Glimepiride; HbA1c: Glycated hemoglobin; Met: Metformin; Sita: Sitagliptin; V: Visit; Wk: Week

- In this subgroup analysis, we compared glycemic outcomes in two subgroups based on baseline HbA1c, i.e, subgroup 1 (S1: 8%-9.5%) and subgroup 2 (S2: $>9.5\%$ -11%) in double-blind period

- Improvement in glycemic outcomes and safety profile of patients were evaluated till Week 16 in this subgroup analysis

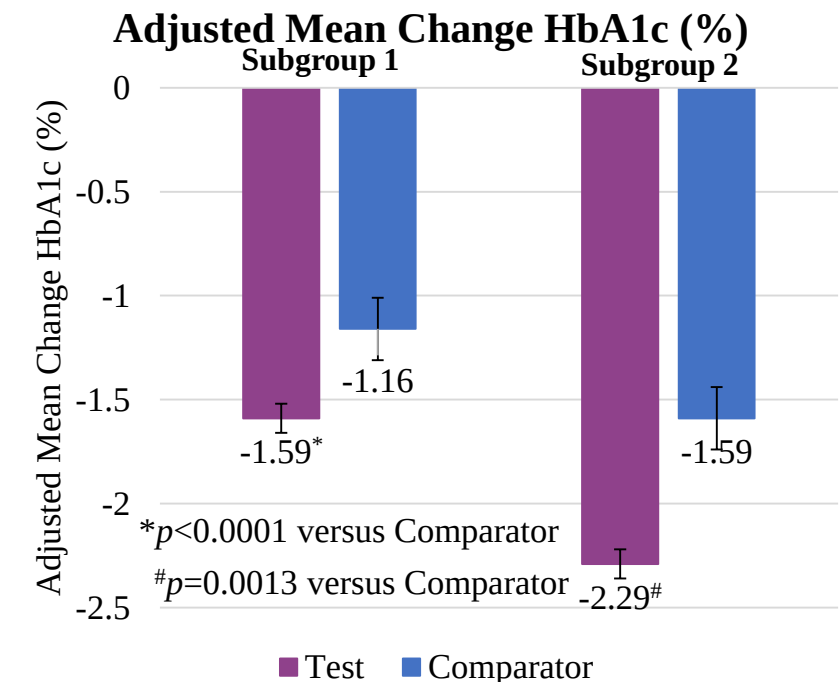
RESULTS

Baseline Demographics

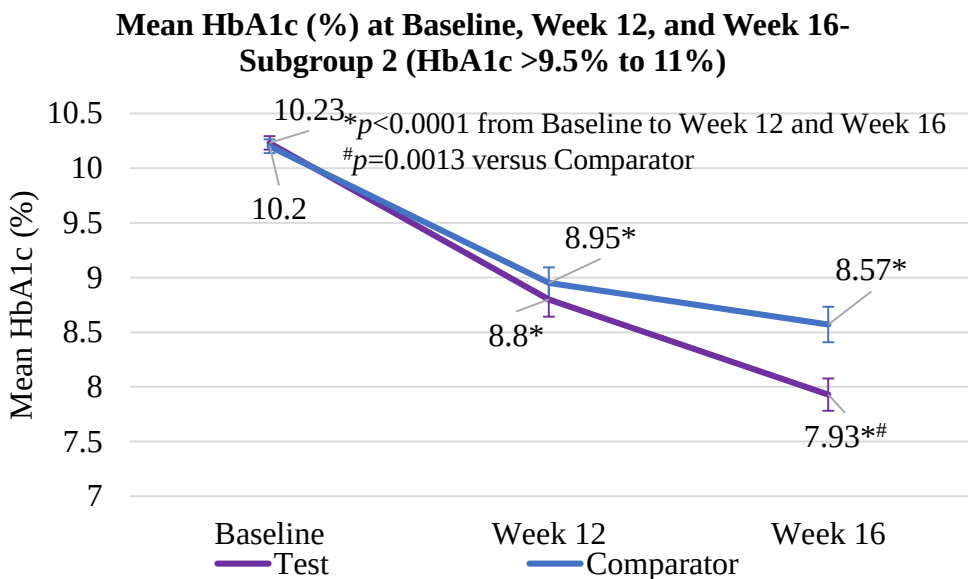
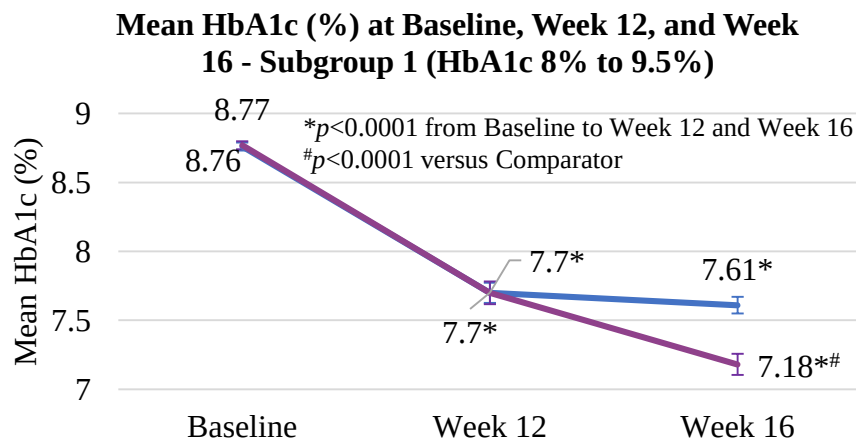
- A total of 286 (137 in Test and 149 in Comparator) and 106 (53 in Test and 53 in Comparator) patients were included in S1 and S2, respectively

Change in HbA1c (%) from Baseline

- In both subgroups, Test arm demonstrated statistically significant improvement in HbA1c (%) as compared to Comparator arm at Week 16 (S1: $p<0.0001$; S2: $p=0.0013$).



- At Weeks 12 and 16, patients in both arms in both subgroups demonstrated significant improvement in HbA1c from baseline ($p<0.0001$ each)



Change in Fasting and Postprandial Blood Glucose from Baseline to Week 16

- Statistically significant reduction in FBG and PPBG from baseline to Weeks 6, 12 and 16 were observed in both arms in both subgroups ($p<0.0001$ each)

Change in Fasting Blood Glucose

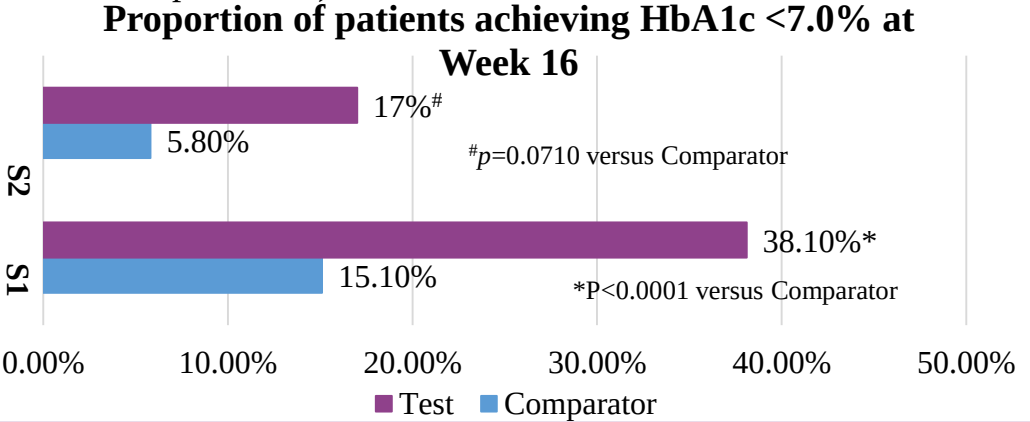
	T	C	T Vs C
Subgroup 1 (HbA1c 8% to 9.5% at Baseline) (Mean $\pm$ SD)			
FBG at Baseline	175.4 $\pm$ 36.12	176.0 $\pm$ 37.76	0.8884
FBG at Week 16	132.4 $\pm$ 25.69	135.9 $\pm$ 26.76	0.2460
CFB	-42.7 $\pm$ 35.11	-39.9 $\pm$ 35.85	
p value	<0.0001	<0.0001	
Subgroup 2 (HbA1c >9.5% to 11% at Baseline) (Mean $\pm$ SD)			
FBG at Baseline	174.2 $\pm$ 43.97	197.5 $\pm$ 50.29	0.0125
FBG at Week 16	137.1 $\pm$ 41.11	155.9 $\pm$ 42.25	0.2286
CFB	-37.1 $\pm$ 47.50	-41.1 $\pm$ 51.55	
p value	<0.0001	<0.0001	

Change in Postprandial Blood Glucose

	T	C	T Vs C
Subgroup 1 (HbA1c 8% to 9.5% at Baseline) (Mean $\pm$ SD)			
PPBG at Baseline	239.5 $\pm$ 46.03	243.8 $\pm$ 54.55	0.4659
PPBG at Week 16	186.3 $\pm$ 37.68	187.3 $\pm$ 35.27	0.9615
CFB	-51.9 $\pm$ 45.17	-55.7 $\pm$ 56.61	
p value	<0.0001	<0.0001	
Subgroup 2 (HbA1c >9.5% to 11% at Baseline) (Mean $\pm$ SD)			
PPBG at Baseline	266.7 $\pm$ 64.73	280.2 $\pm$ 74.86	0.3237
PPBG at Week 16	197.7 $\pm$ 48.19	209.2 $\pm$ 50.09	0.4668
CFB	-69.0 $\pm$ 64.54	-70.7 $\pm$ 67.86	
p value	<0.0001	<0.0001	

Proportion of Patients Achieving HbA1c <7.0%

- In S1, proportion of patients achieving HbA1c $<7.0\%$ was statistically significant in test versus comparator (S1:38.1% vs 15.1%, $p<0.0001$) at Week 16



Overall Adverse Event Summary till Week 16

- Safety profile was comparable between both arms in both subgroups, and the treatments were safe and well tolerated
- Out of 36 TEAEs (Test=17; Comparator=19) in S1, 20 were drug-related (10 in each arm), and out of 31 TEAEs (Test=14; Comparator=17) in S2, 8 were drug-related (Test=2 and Comparator=6)
- The most common TEAEs were headache, vomiting, diarrhea, and nausea in S1 and hypoglycemia and pyrexia in S2
- In this analysis, no serious AEs were reported and none of the patients were discontinued due to AEs and all TEAEs were resolved
- None of the patient required rescue medications during the study

DISCUSSION AND CONCLUSION

- Irrespective of the HbA1c value at Baseline, Test demonstrated statistically superior efficacy as compared to the Comparator and a statistically significant reduction in glycemic parameters were observed from Baseline to Week 16 in both the treatment arms
- Thus, the synergistic mechanism of action of Sitagliptin in combination with Metformin demonstrated an optimum glycemic control with a lower dose of Glimepiride in patients who were previously on a higher dose of glimepiride
- In S1 patients, statistically significant proportion of patients achieved glycemic control (HbA1c $<7.0\%$) at Week 16 in Test vs Comparator arm, while in S2 patients, higher proportion of patients achieved glycemic control in Test vs Comparator arm
- Safety profile was comparable between both the arms and both the study treatment were safe and well tolerated

Abbreviations - C: Comparator; CFB: Change from Baseline; DPP4: Dipeptidyl peptidase-4; GLP1: Glucagon-like peptide 1; SD: Standard deviation; T: Test